

Global Semiconductor Supply Chain Risk Assessment

Cross-Company Intelligence Report — 2024

Covering: Apple, TSMC, Nvidia, ASML, Samsung

Executive Summary

This report analyses the interconnected supply chain dependencies among the world's leading semiconductor companies and the technology companies that depend on their output. The central finding is that a small number of chokepoints — most critically ASML's EUV lithography monopoly and TSMC's advanced foundry dominance, both concentrated in geopolitically sensitive regions — create systemic risk for the entire global technology sector.

The dependency chain flows as follows: ASML (Netherlands) is the sole global supplier of EUV lithography equipment. TSMC (Taiwan) is ASML's largest customer and uses EUV machines to manufacture chips at 3-nanometer and below process nodes. Apple (USA) relies on TSMC as the sole manufacturer of all Apple Silicon chips, which generate revenue across iPhone, Mac, and iPad product lines totalling USD 391 billion in fiscal year 2024. Nvidia (USA) relies on TSMC to manufacture its H100 and Blackwell GPUs, which generated USD 47.5 billion in data centre revenue in fiscal year 2024.

1. The Four-Layer Dependency Chain

The most critical insight of this report is that Apple and Nvidia — two of the world's most valuable companies — are both exposed to the same upstream supply chain risks through their shared dependence on TSMC as a sole-source manufacturer. And TSMC itself is dependent on ASML as its sole supplier of EUV lithography equipment. This creates a four-layer cascade risk:

Layer	Company	Dependency	Risk if Disrupted
Layer 1	ASML	Sole EUV manufacturer	Global advanced chip production halts
Layer 2	TSMC	Sole advanced foundry for Apple & Nvidia	Apple + Nvidia lose chip supply
Layer 3	Apple	TSMC sole-source for all Apple Silicon	iPhone, Mac, iPad production halts
Layer 3	Nvidia	TSMC primary source for all GPU lines	H100/B200 GPU supply halts
Layer 4	Cloud	AWS, Azure, GCP depend on Nvidia GPUs	AI infrastructure capacity constrained

2. Taiwan Geopolitical Risk — Quantified Impact

A disruption to TSMC's Taiwan operations — whether caused by military conflict, a naval blockade, severe earthquake, or extended power grid failure — would have immediate and severe financial consequences for Apple and Nvidia.

Company	Annual Revenue at Risk	Mechanism	Recovery Timeline Estimate
Apple	USD 391B (100% of sales)	Loss of all Apple silicon from TSMC; no alternative foundry	3-5 years to qualify alternative
Nvidia	USD 60.9B (95% of sales)	H100/H200 manufactured at TSMC N4/N4P	2-3 years to qualify Samsung
TSMC	USD 69.3B revenue loss	Direct loss of all manufacturing revenue	TSMC is the disrupted entity
ASML	~35% revenue at risk	Loss of TSMC as largest customer	Could redirect to Samsung/Intel

3. ASML Export Control Risk — Second-Order Effects

Export control restrictions on ASML's EUV and advanced DUV systems have already limited China's ability to manufacture chips below 7 nanometers. However, export controls also create risks for the Western semiconductor ecosystem. If ASML's China revenue — approximately 26% of total revenue in FY2023 — is eliminated through further restrictions, ASML's profitability and R&D; investment capacity could be impaired, potentially slowing the development of next-generation High-NA EUV systems that TSMC and Intel require for their sub-2-nanometer process nodes.

The second-order effect of this slowdown would reach Apple and Nvidia: delayed availability of 2nm and below process nodes at TSMC would slow Apple's chip performance roadmap (affecting iPhone and Mac competitiveness) and delay Nvidia's next-generation GPU architectures beyond Blackwell, potentially allowing competitors such as AMD and custom silicon from Google, Amazon, and Microsoft to close the performance gap.

4. Samsung's Role — Competitor and Alternate Supplier

Samsung Electronics occupies a unique dual role in this supply chain. Samsung Foundry competes with TSMC for advanced chip manufacturing contracts, making Samsung a potential alternative supplier for Nvidia's GPU production. Nvidia has publicly stated it is evaluating Samsung Foundry for manufacturing certain GPU product lines to reduce its concentration risk at TSMC.

Simultaneously, Samsung is a supplier to Apple — providing OLED display panels used in iPhone models and DRAM and NAND flash memory components. This means Samsung is both a competitor to Apple in smartphones and a critical component supplier to Apple, a relationship that creates complex commercial tensions. Samsung also purchases ASML EUV systems for its own foundry operations, making Samsung a customer of ASML, a customer of Apple (as a consumer buying iPhones), and a supplier to both Apple and Nvidia.

5. Key Multi-Hop Risk Scenarios for Analysts

Scenario A: ASML EUV export ban expands to all markets

If geopolitical pressure forced ASML to halt all EUV shipments globally (including to TSMC), advanced chip production at TSMC would decline as existing machines age and fail. Within 2-3 years, Apple's A-series and M-series chip performance would plateau, Nvidia's GPU roadmap would stall, and cloud AI infrastructure expansion would slow globally. ASML's own revenue would collapse and R&D investment would be impaired.

Scenario B: Taiwan military conflict — 90-day scenario

A 90-day disruption to TSMC's Taiwan operations would exhaust Apple's chip inventory within 4-6 weeks (Apple does not maintain large semiconductor stockpiles). iPhone production would halt, causing Apple to miss an estimated USD 50-70 billion in quarterly revenue. Nvidia GPU shipments would stop, causing hyperscale cloud providers to delay AI infrastructure expansion. Global chip prices would spike across all product categories.

Scenario C: US-Netherlands export control rift

If the Netherlands reduced cooperation with US export control policy, ASML could resume EUV exports to China. Chinese chipmakers (primarily SMIC and CXMT) would gain access to sub-7nm production capability within 3-5 years. This would intensify competition for TSMC and reduce Apple's chip manufacturing cost advantage, while increasing competitive pressure on Nvidia from Chinese AI chip startups such as Huawei's Ascend division.